BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Random Numbers and Combined Constructs

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does SELECT specify in an SQL query?

2. What does a higher CPU clock speed mean?
   A) More installed cores
   B) More bytes per pixel
   C) More cycles per second
   D) More storage per file

3. "Larger cache increases the number of installed CPU cores." Correct this. Explain why.

4. CPU A: 4 GHz, two cores, 2 MB cache. CPU B: 3 GHz, eight cores, 8 MB cache. Why is neither fastest for every task?

Today's objectives

Explain the core ideas of random numbers and combined constructs using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Random integer
A pseudorandom function can select a whole number from a stated inclusive range when specified.

Sequence
Statements execute one after another unless control flow changes.

Selection
A condition decides whether a branch runs.

Iteration
A loop repeats a process while or until an appropriate stopping condition is met.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Random integer
A pseudorandom function can select a whole number from a stated inclusive range when specified.

Sequence
Statements execute one after another unless control flow changes.

Selection
A condition decides whether a branch runs.

Iteration
A loop repeats a process while or until an appropriate stopping condition is met.

I Do - Worked model

Trace a dice game: roll=6; if roll>=5 print('Win'), else print('Try again').

- Identify the given roll 6.
- Evaluate 6 >= 5 as true.
- Choose the winning branch.
- The output is Win.

We Do - Guided practice

1. A simulated roll is 3 and the program wins only on 6. Predict the output.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the bonus-message output when the dice roll is six.

```
roll = random(1,6)
if roll == 6 then
print(...)
else
print("Keep trying")
endif
```

You Do - Independent exam questions

Q2. Explain the purpose of random number generation in a dice simulation. [2 marks]

Q3. State two programming constructs in a game with a loop and IF statement. [2 marks]

Q4. Why should a game test both the minimum and maximum permitted dice rolls? [2 marks]

Exit Ticket - True or False?

A condition can determine which branch of a program runs.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. The fields (columns) to return.

2. C - More cycles per second

3. A larger cache holds more often-used data and instructions. This can reduce slower memory access.

4. A has more cycles per second, which may help one task. B has more cores for parallel work and more cache to reduce slow memory access. Speed also depends on the task and CPU design.

I Do - Explain the model

Identify the given roll 6.

Evaluate 6 >= 5 as true.

Choose the winning branch.

The output is Win.

We Do - Model answers

1. The output is Try again.

Code First - Answer

```
roll = random(1,6)
if roll == 6 then
print("Bonus!")
else
print("Keep trying")
endif
```

Expected output: Bonus! if roll is 6; otherwise Keep trying

You Do - Self-mark

Q2: It models different possible outcomes rather than a fixed predetermined roll.

Q3: Iteration and selection.

Q4: They are boundaries of the expected range and may expose off-by-one errors.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.